Report on CIVENG Hackathon 2024

The **CIVENG Hackathon 2024** was successfully conducted with the objective of fostering innovation, creativity, and collaboration among civil engineering students under Institution of engineers Students chapter. The hackathon, themed on sustainable solutions and technological advancements in civil engineering, saw enthusiastic participation from multiple batches. The event aimed to encourage students to develop solutions that align with sustainable development goals, enhancing practical knowledge and problem-solving skills. The Department of Civil Engineering at Vasavi College of Engineering, Hyderabad, conducted the CIVENG Hackathon 2024, an exciting competition aimed at fostering innovation and sustainable development in the field of civil engineering. Sponsored by esteemed alumni, Mr. Aditya Mokkapati (Batch of 2010) and Mr. Arunav Mokkapati (Batch of 2007) on **13.11.2024**.This hackathon invites the brightest minds to come together, brainstorm, and design cutting-edge solutions to real-world challenges.

Themes of the Hackathon:

Participants can choose to work on one of the following thought-provoking themes:

1. **Sustainable Infrastructure Development**: As the world grapples with environmental concerns, sustainable infrastructure is becoming increasingly important. This theme challenges students to develop models that focus on creating eco-friendly, resilient, and cost-effective infrastructure solutions.
2. **Future of Construction Technologies**: Construction is evolving rapidly, with technologies like AI, automation, and advanced materials reshaping the industry. Under this theme, participants will explore innovative construction techniques and emerging technologies that can revolutionize how buildings and infrastructure are designed and built.
3. **Rural Development**: With rural areas being pivotal for the socio-economic growth of the country, this theme encourages students to create sustainable solutions that enhance the quality of life in rural communities, addressing issues such as access to basic amenities, infrastructure, and resources.

Event Details:

- **Date**: November 13, 2024
- **Venue**: Environmental engineering lab, Civil Engineering Department, Vasavi College of Engineering
- **Total Participating Teams**: 10
- **Participants**: Students from 2nd, 3rd, and 4th year B.E. Civil Engineering

Each team comprised students from different years, fostering cross-year collaboration. The projects addressed real-world challenges in civil engineering, focusing on sustainability, waste management, renewable energy, and innovative construction techniques.

























Project Presentations and Evaluation

The projects were evaluated based on the **Hackathon Rubrics** which included parameters like Innovation and Creativity, Technical Feasibility, Sustainability, Presentation, and Scalability. The judging panel consisted of experienced faculty members and industry experts.

Below are the details of the participating teams and their project titles

Batch No	Hall ticket no	Name of the student	Year of study	Title of the project
1	1602-21-732-008	KETHAVATH. BADHU	4/4	Waste to wealth
	1602-22-732-303	KOPPULA KEERTHANA	3/4	
	1602-23-732-007	UPPALA BHARATH	2/4	
2	1602-21-732-306	ABDUL. JUNAID	4/4	Radiant Cooling system
	1602-22-732-307	Mohammed Mutahhir Ali Khan	3/4	
	1602-23-732-001	SANIYA NAUSHIN	2/4	
3	1602-20-732-038	SINGARI. SRIVARDHAN	4/4	Transformation of a village into sustainable village
	1602-22-732-308	Meganath Charugundla	3/4	
	1602-23-732-003	DHARAVATH ABHINAYA	2/4	
4	1602-21-732-059	BANOTHU. VIJAY	4/4	Sand replacement by quarry dust by various proportions
	1602-22-732-302	Alli Sujatha	3/4	
	1602-23-732-308	KONDAMEEDHI MADHAVAN	2/4	
5	1602-21-732-305	K. SAILAXMI	4/4	Sustainable porous pavements using

	1602-22-732-301	Chelamalla Lavanya	3/4	recycled aggregates and geopolymers
	1602-23-732-305	KATTERAPAKA SOWMYA	2/4	
6	1602-21-732-022	AITHA. KOUSHIK	4/4	Zero emission building
	1602-22-732-031	Sri Hari Chandra Katam	3/4	
	1602-23-732-027	MUGDE NAGAJYOTHI	2/4	
7	1602-21-732-026	S. T NIHAL	4/4	Self sustaining bus stops
	1602-22-732-313	DHANAVATH VIKRAM RATHOD	3/4	
	1602-23-732-018	MUSKAN KAUSAR	2/4	
8	1602-21-732-037	SIDDAM. SAI SAKETH	4/4	A smart system for segregated waste collection and energy recovery
	1602-23-732-009	VISTARAKULA CHARAN	2/4	
	1602-23-732-002	BEERAVELLI ABHINAV REDDY	2/4	
9	1602-21-732-009	GADDAMEEDHI. BHARATH KUMAR	3/4	Automated irrigation system
	1602-23-732-010	DESHAPAGA CHETAN	2/4	
	1602-23-732-033	RAMAKANTH KAKULAPATI	2/4	
10	1602-23-732-056	DONIPARTHI VENKATESH TEO	2/4	Tornado turbine
	1602-23-732-307	PAGIDIPALLI DIVIJ KUMAR	2/4	
	1602-23-732-015	KANTHA GOUTHAMI	2/4	

Judges

1. Dr.C.Mohanlal, Associate Professor, CED
2. Mr.G.Raghavendra, Associate Professor, CED
3. Dr.Swarnadeepa Chakraborthy, Assistant Professor, CED

Event Coordinator: Dr.K.Jayasree, Associate Professor, CED

Special Recognition: Best Projects

1. Team 8 - *"A Smart System for Segregated Waste Collection and Energy Recovery"*

- **Team Members**:
  - **Siddam Sai Saketh** (4th Year)
  - **Vistarakula Charan** (2nd Year)
  - **Beeravelli Abhinav Reddy** (2nd Year)
- **Highlights**:
  - Developed a smart waste management system leveraging IoT technology to segregate waste and convert organic waste into energy.
  - The project focused on sustainability by promoting efficient waste management and energy recovery, reducing landfill impact.
  - Commended for its practical feasibility and potential real-world impact.

2. Team 10 - *"Tornado Turbine"*

- **Team Members**:
  - **Doniparthi Venkatesh Teo** (2nd Year)
  - **Pagidipalli Divij Kumar** (2nd Year)
  - **Kantha Gouthami** (2nd Year)
- **Highlights**:

- Designed an innovative wind turbine system inspired by the dynamics of tornadoes to optimize energy capture in low-wind conditions.
- The project showcased creativity in harnessing renewable energy, making it highly relevant for sustainable power generation.
- Recognized for its technical soundness and potential scalability in real-world applications.

3. Team 5: Consolation and Appreciation Prize

- **Team Members**:
    - K. SAILAXMI(4th Year)
    - Chelamalla Lavanya (3rd Year)
    - KATTERAPAKA SOWMYA(2nd Year)

Key Achievements

- **Best Project Awards**:
    - **Team 8** - *Smart Waste Management System*
    - **Team 10** - *Tornado Turbine*
- **Most Innovative Solution**: Team 8 for their cutting-edge IoT integration.
- **Technical Excellence**: Team 10 for the unique approach in renewable energy capture.

Conclusion

The **CIVENG Hackathon 2024** was a resounding success, encouraging students to think beyond traditional civil engineering practices and focus on sustainable and innovative solutions. The projects demonstrated the students' capability to apply their academic knowledge to solve practical problems, contributing to the broader goals of sustainable development.

Dr.K.Jayasree
Associate Professor, CED